

DATA CLEANING & QUALITY ASSESSMENT

Global Life Expectancy Paradox Dataset

Comparison of the dataset before and after cleaning

1. Executive Summary

Purpose. This report documents the data-quality condition of the Global Life Expectancy Paradox dataset before cleaning and compares it with the final cleaned version. The review focuses on missing values, duplicate records, consistency of the dataset structure, and the effect of the cleaning process.

Overall finding. The original dataset contains 204 records and 20 variables. The final cleaned dataset contains 192 unique records and 20 variables. The 154 missing cells were filled through median/mode imputation, and 12 exact duplicate rows were removed.

2. Dataset Profile

Measure	Before cleaning	After cleaning
Rows	204	192
Columns	20	20
Missing cells	154	0
Duplicate rows	12	0
Numeric columns	17	17
Column structure	20 columns	20 columns

3. Dataset Before Cleaning

The original workbook is structurally usable, but several fields contain missing observations. The missingness is concentrated in a small number of variables, with CO₂ emissions accounting for the largest share. The original file also contains 12 exact duplicate rows, which were removed during final cleaning.

Variable	Missing values	% of rows
co2_emissions	59	28.9%
homicide_rate	23	11.3%
secondary_school_enrollment_male	14	6.9%
secondary_school_enrollment_female	14	6.9%
unemployment	10	4.9%
infant_mortality	8	3.9%
life_expectancy_male	6	2.9%
life_expectancy_female	6	2.9%
fertility	5	2.5%
forested_area	4	2.0%

internet_users	2	1.0%
gdp_growth	1	0.5%
iso2	1	0.5%
gdp_per_capita	1	0.5%

Missingness totals 154 cells across the dataset. The highest count occurs in co2_emissions (59 missing values), followed by homicide_rate (23). There are also 12 exact duplicate rows, meaning the raw file contains repeated country records. These 12 duplicate rows were removed in the final cleaned dataset.

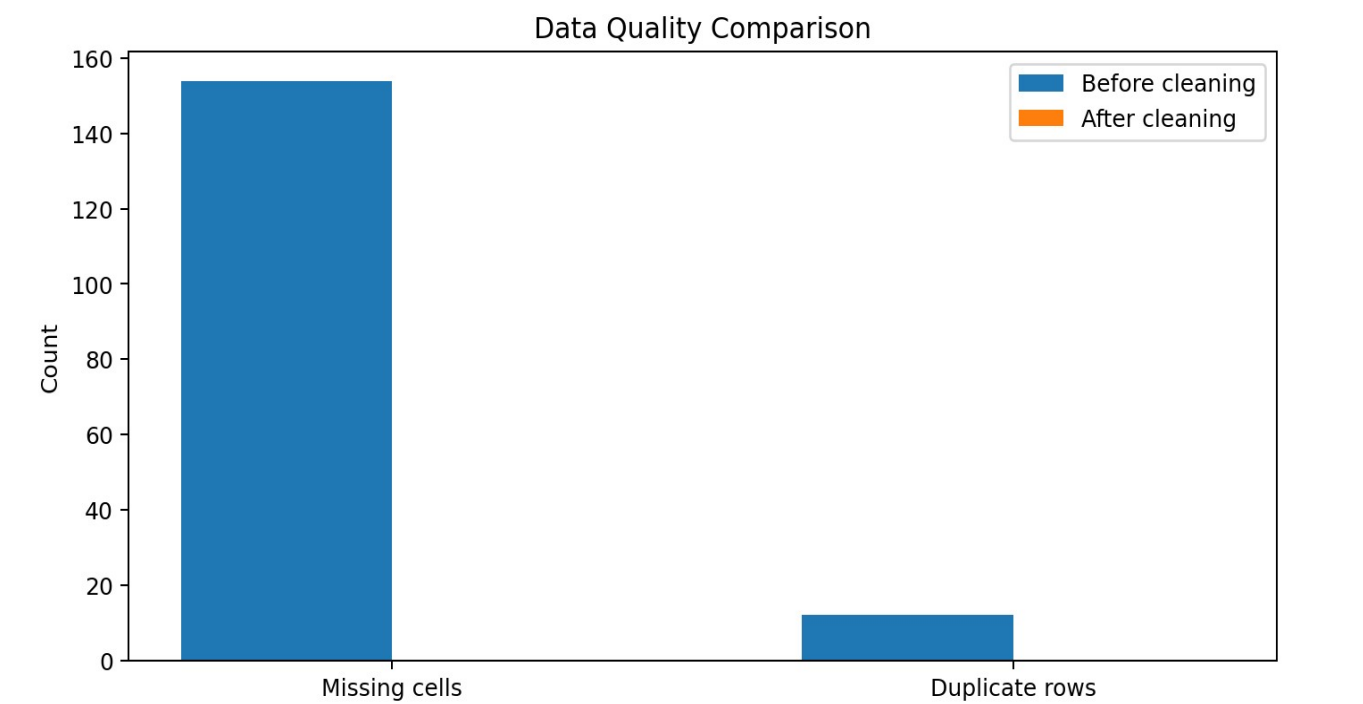


Figure 1. Comparison of key data-quality indicators before and after cleaning.

4. Cleaning Performed

The supplied cleaned dataset was examined against the original workbook to identify what changed. The cleaning process addressed missing values and then removed exact duplicate rows, resulting in 192 unique records while retaining all 20 variables.

Variable	Values filled	Method identified
iso2	1	Mode
life_expectancy_male	6	Median
life_expectancy_female	6	Median
gdp_per_capita	1	Median
gdp_growth	1	Median
unemployment	10	Median
infant_mortality	8	Median
fertility	5	Median
homicide_rate	23	Median
internet_users	2	Median
secondary_school_enrollment_female	14	Median
secondary_school_enrollment_male	14	Median
co2_emissions	59	Median
forested_area	4	Median

Duplicate rows	12	Exact duplicate removal
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For the numeric variables, the replacement values in the cleaned file match the corresponding median of the non-missing observations in the original dataset. The missing ISO2 value was filled with the categorical mode. After imputation, 12 exact duplicate rows were removed, retaining one copy of each duplicated record. This produces a complete, duplicate-free analysis dataset.

5. Dataset After Cleaning

The final cleaned file contains no blank cells across the 20 variables and contains no exact duplicate rows. The column names and data types remain consistent with the original dataset, while the row count decreases from 204 to 192 because 12 duplicate rows were removed. This makes the file more suitable for statistical analysis, visualization, correlation analysis, and subsequent modelling.

Quality indicator	Before	After	Change
Missing cells	154	0	154 removed
Duplicate rows	12	0	12 removed
Rows	204	192	12 duplicate rows removed
Columns	20	20	No change

6. Effect on the Dataset

Median imputation removes gaps without deleting observations, but it can slightly alter summary statistics. Removing exact duplicate rows can also change summary statistics because repeated records no longer receive extra weight. The table below provides selected descriptive statistics for comparison.

Variable	Mean before	Mean after	Median before	Median after
life_expectancy_male	69.82	69.64	70.65	70.55
life_expectancy_female	74.65	74.51	76.70	76.55
gdp_per_capita	15,888.43	15,536.08	5,930.10	5,862.35
unemployment	7.30	7.07	5.40	5.40
infant_mortality	21.94	21.95	14.35	14.35
fertility	2.74	2.75	2.30	2.30
homicide_rate	6.83	6.54	2.50	2.50
internet_users	54.72	54.11	58.95	57.60
co2_emissions	224.64	171.09	31.30	31.30
forested_area	31.50	32.10	31.55	31.65

7. Quality Assessment

Completeness: Improved substantially. All 154 missing cells present in the original file have been populated in the final cleaned dataset.

Consistency: The same 20 variables are retained, and the data types remain appropriate for analysis.

Record retention: The final dataset contains 192 unique rows. Twelve exact duplicate rows were removed; no observations were removed because of missing values.

Duplicate control: Fully addressed. The final cleaned dataset contains 0 exact duplicate rows after removing 12 duplicate rows.

Analytical readiness: The final cleaned file is substantially more suitable for exploratory data analysis and modelling because it contains no missing cells and no exact duplicate rows.

8. Recommendations

- Maintain an exact-duplicate check as a standard step in future versions of the cleaning workflow; 12 duplicate rows were removed in this version.
- Verify whether duplicate country records are intentional or accidental before deletion in future raw datasets.
- Document the median/mode imputation rule and duplicate-removal rule in the project methodology so the cleaning process is reproducible.
- For variables with relatively high missingness, especially `co2_emissions`, consider a sensitivity check to determine whether median imputation affects correlations or regression results.
- Retain the original workbook as the raw-data version and use the final cleaned file as the analysis version.

9. Conclusion

The comparison shows a clear improvement in completeness and record quality between the original and final cleaned versions of the Global Life Expectancy Paradox dataset. The 154 missing cells were eliminated through value imputation, and 12 exact duplicate rows were removed. The final dataset therefore contains 192 unique rows and 20 variables, with 0 missing cells and 0 exact duplicate rows. With the imputation and duplicate-removal rules documented, the dataset provides a stronger and more reliable foundation for further statistical, exploratory, and modelling analysis.